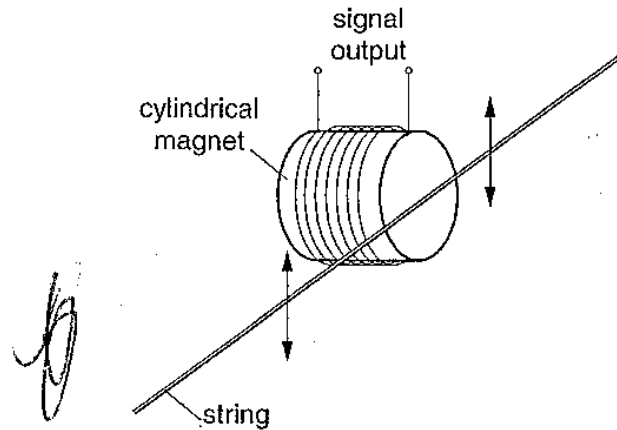


Q1: NOV 2001 P2

- 6 The pick-up on an electric guitar produces an electrical signal from the vibrations of the guitar strings. The pick-up consists of a small coil of insulated wire wound round a small cylindrical bar magnet as illustrated in Fig. 6.1.

**Fig. 6.1**

The strings of the guitar are made of steel. When a string vibrates, an electrical signal is generated between the terminals of the coil.

- (a) (i) State Faraday's law of electromagnetic induction.

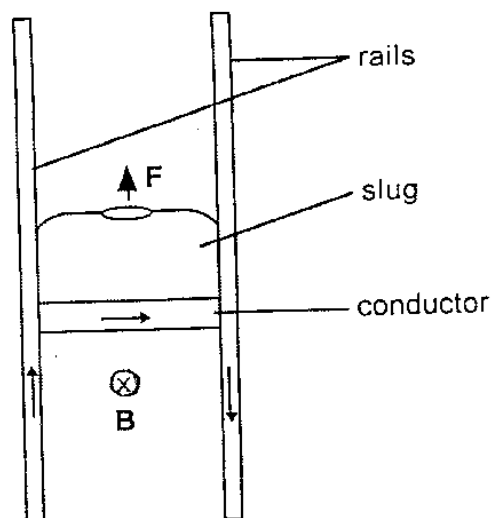
.....

.....

- (ii) Use Faraday's law to explain why an electrical signal is generated.

Q2: NOV 2002 P2

- 5 A force acts on a current-carrying conductor placed in a magnetic field. Electromagnetic propulsion devices such as the rail gun (refer to Fig. 5.1) use this principle to propel objects such as a slug.



4.5-4.6 ELECTROMAGNETISM AND INDUCTION P2, P3&P5 QUESTIONS

Compiled by : *R. Nuzo*

A current that flows through the conducting rails as shown in Fig.5.1 will give rise to a magnetic field **B** that is directed into the page. The conductor falls within this field and thus experiences a force **F**.

- (a) Define the *tesla*, the unit of magnetic flux density.

[1]

- (b) State and explain how the direction of the force **F** on the conductor is predicted.

[2]

- (c) (i) Draw on Fig. 5.1 the magnetic field pattern due to the conducting rails.

[2]

- (ii) Hence draw arrows showing the forces acting on the rails.

[1]

- (iii) The forces in (ii) above may present problems. State one possible problem and explain how it can be taken care of.

[2]

- (d) State two factors that the force **F** depends on.

[2]

- (e) Suggest one use for such a railgun other than for military purposes.

[1]

Q3: NOV 2004 P2

- 6 (a) A rectangular coil of N turns and area A , is placed between two poles of a magnet providing a magnetic field of flux density B (see Fig. 6.1).

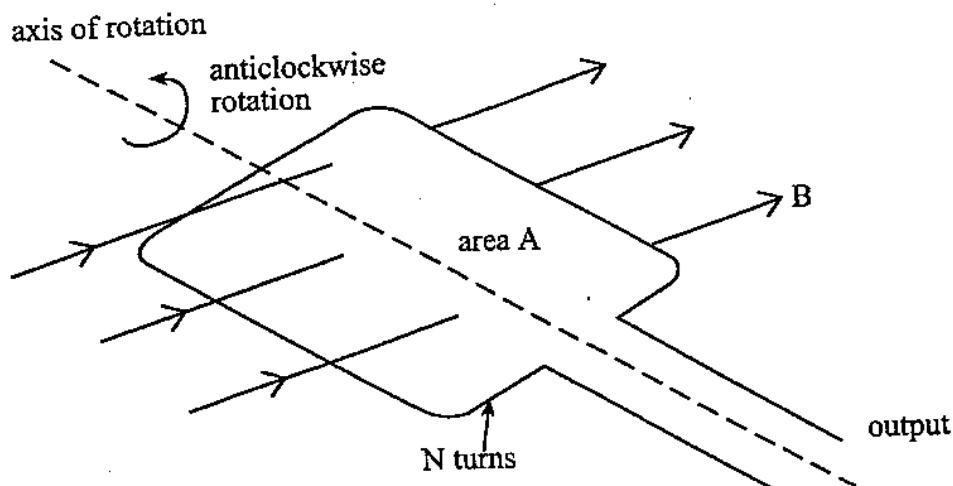


Fig. 6.1

- (i) If the angle between the plane of the coil and the direction of the field is θ , show that the total flux Φ cut by all the turns of the coil is given by

$$\Phi = BAN \sin \theta.$$

[3]

- (ii) Hence deduce an expression for the induced e.m.f in the coil.
[Hint: if coil turns through angle θ in time t then the angular frequency ω is given by $\omega = \frac{\theta}{t}$]

[2]

- (b) State **three** ways in which the magnitude of the e.m.f can be increased.

[3]

Q4:NOV 2009 P2

- 6 (a) Give a statement of the

- (i) *Lenz's law*,

- (ii) *Faraday's law*.

[2]

- (b) State with a reason which of the two laws involves conservation of energy.

[2]

Q5:JUN 2010 P2

- 6 (a) The e.m.f. induced in a coil is given by $E = -\frac{Nd\Phi}{dt}$

(i) Give the meanings of the symbols N and ϕ .

(ii) State the significance of the minus sign.

[3]

- (b) Fig. 6.1 shows a beam of electrons passing undeflected through a magnetic field of flux density 0.02 T perpendicular to an electric field.

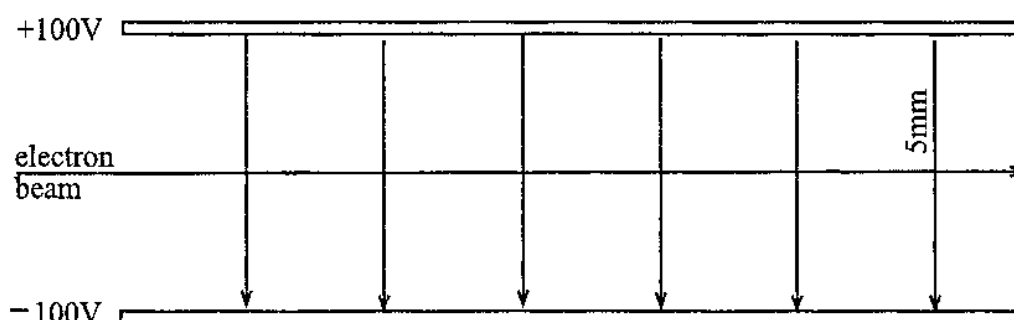


Fig. 6.1

Calculate

- (i) the force due to the electric field on an electron,

force = _____

- (ii) the speed of electrons on leaving the crossed filed.

speed = _____

[4]

Q6: NOV 2012 P2

- 6 (a) Define *magnetic flux density*.

 _____ [1]

- (b) Section AB of a rectangular current-carrying conductor, of length 7.0 cm, is in a magnetic field, of field strength 0.021 T, as shown in Fig. 6.1.

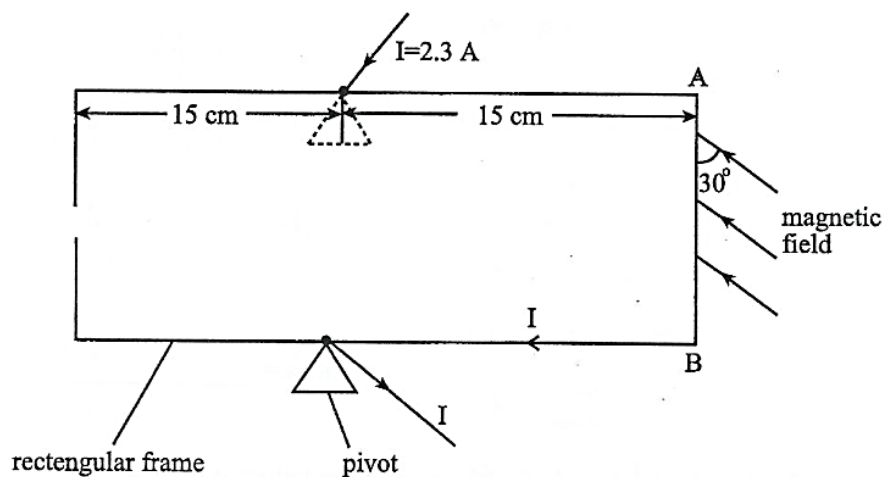


Fig. 6.1

A current of 2.3 A is allowed to pass through the conductor.

- (i) Describe the effect of passing the current through part AB.

- (ii) Calculate the magnitude of the force experienced by the frame due to the current.

force = _____

- (iii) Deduce a value of mass that has a weight equal to the force calculated in (b) (ii).

mass = _____

- (iv) Explain the physical significance of the value of mass deduced in (b) (iii).

Q7:NOV 2013 P2

5

Fig. 5.1 shows a sliced copper plate undergoing vertical oscillations. An electromagnet is placed near the centre of the sliced copper plate. The electromagnet is then switched on.

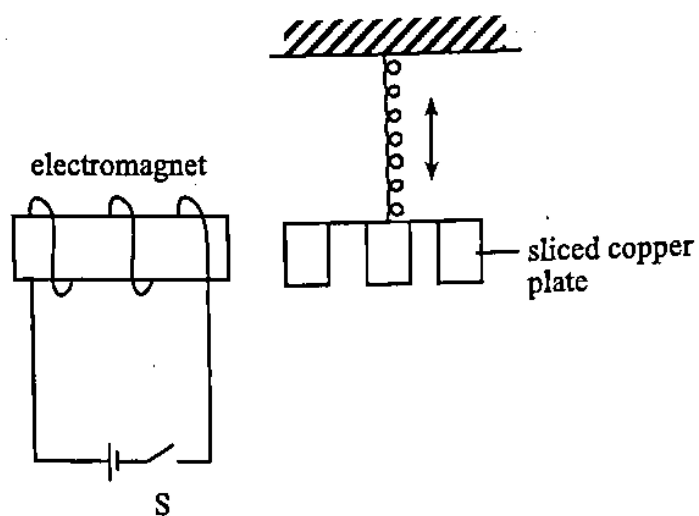


Fig. 5.1

- (i) Sketch and explain a graph showing the variation of displacement with time of the copper plate.

[3]

- (ii) Name and state the law that explains this phenomenon of electromagnetic induction.

[2]

- (iii) Explain how Fig. 5.1 can be modified to demonstrate heavy damping.

[2]

Q8:JUN 2015 P2

- 5 (b) A 5.0 cm long copper wire carrying a current of 2.0 A is placed in a magnetic field. The wire experiences a force of 4.0×10^{-4} N when inclined at 50° to the field.

Calculate the magnetic flux density, B , of the field.

$$B = \underline{\hspace{2cm}} \quad [2]$$

- (c) Describe how a calibrated Hall probe can be used to measure flux density.

[3]

Q9:JUN 2018 P2

- 5 (a) State

- (i) Faraday's law,

- (ii) Lenz's law.

[2]

- (b) Coil X connected to a battery and a switch is placed near coil Y, which is connected to a galvanometer, as shown in Fig.5.1.

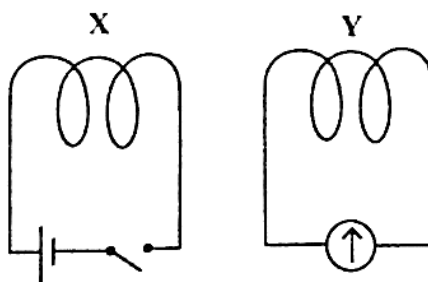


Fig. 5.1

Explain why when the switch is closed

1. the galvanometer deflects momentarily,

2. coil Y moves to the right,

3. the galvanometer reads zero when the switch is kept closed.

[6]

Q10:2018 SPEC P2

6. (a) State Faraday's law of electromagnetic induction.

[1]

- (b) Fig. 6.1 shows a lump of soft solders at the centre of a coil. The coil is connected to a signal generator.

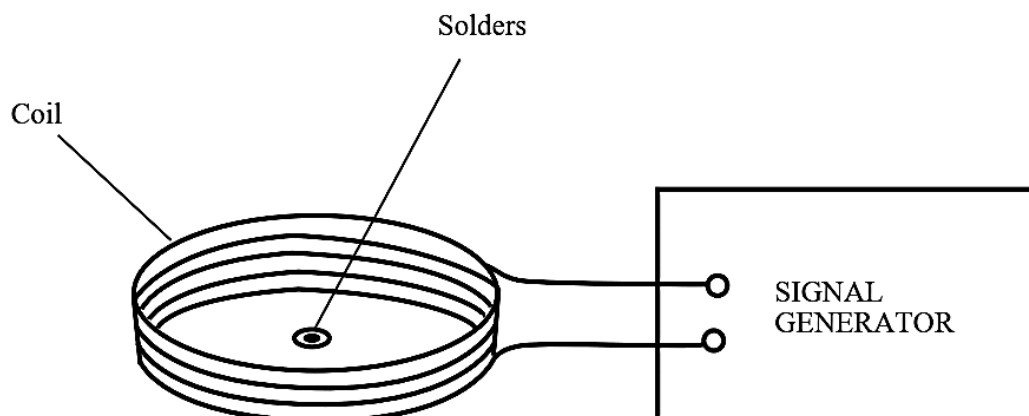


Fig. 6.1

Describe and explain what happens to the solder when the a.c. frequency is increased to a high value.

[5]

Q11: NOV 2007 P3

- 5 (a) Define *magnetic flux* and the *weber*. [2]
- (b) Fig. 5.1 shows a square coil of sides 50.0 cm moving at a steady speed of 0.25 ms^{-1} through a **uniform** vertical magnetic field of flux density 0.050 T.

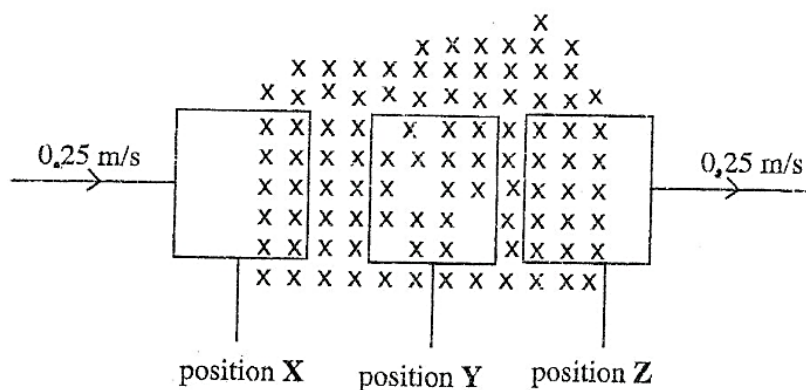


Fig. 5.1

4.5-4.6 ELECTROMAGNETISM AND INDUCTION P2, P3&P5 QUESTIONS

Compiled by: R. Nuzo

- (i) Calculate the magnetic flux linking the coil when the coil is at position Y.
- (ii) Sketch graphs to show
- 1 how the magnetic flux through the coil varies with time,
 - 2 how the induced e.m.f. in the coil varies with time as the coil moves from X to Z.
- (iii) Calculate the induced current when the coil has moved into the field for 1.0 s given that the coil has a resistance of $5\ \Omega$. [10]
- (c) Describe how a Hall probe can be used to measure flux density. [6]
- (d) A resistor of $6.0\ \Omega$ is connected to a secondary coil of a transformer with 1 200 turns. The primary coil of 2 400 turns is connected to a 240 V ac supply. Calculate the current through the $6.0\ \Omega$ resistor. [2]

Q12:JUN 2007 P3

- 7 (a) State the laws of electromagnetic induction. [2]
- (b) A conductor is placed in the magnetic field. The flux of the field varies with time as shown in Fig. 7.1.

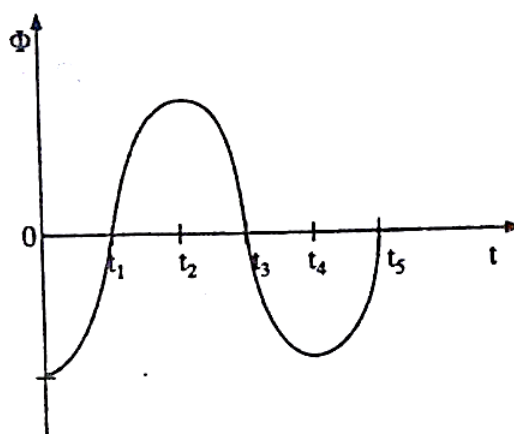


Fig. 7.1

- (i) Sketch a graph to show how the e.m.f. across the conductor varies with time.
- (ii) Explain the sketch in relation to Fig. 7.1. [5]

Q13: NOV 2006 P3

- 4 (c) (i) Define magnetic field strength in terms of force.
- (ii) A wire of length, L , carrying a current, I , at right angles to a magnetic field of flux density, B experiences a force, F . Deduce the angle, θ , between the wire and the field for which the force is
1. $\frac{F}{3}$,
 2. 0.
- (d) (i) Fig. 4.2 shows a solenoid carrying a current and a wire AB passing through the solenoid.

[5]

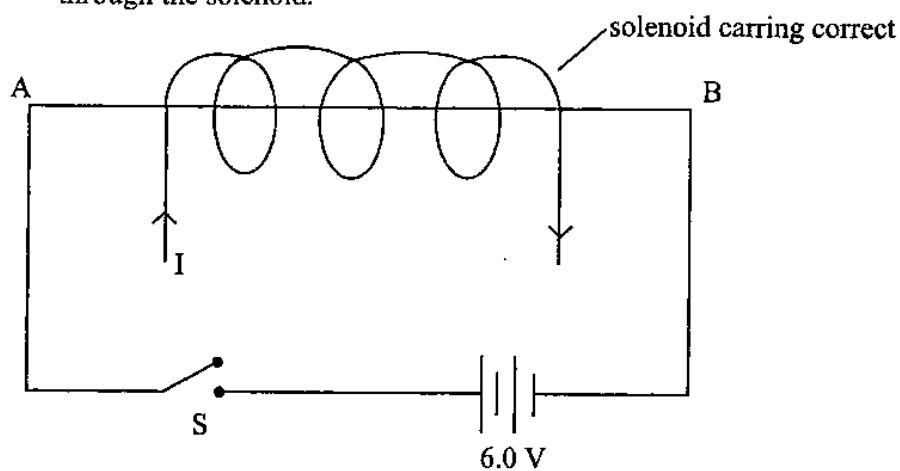


Fig. 4.2

State and explain the effects on wire AB of closing the switch S.

- (ii) In Fig. 4.3 wire P is connected to a sensitive galvanometer and wire Q is connected to a switch and a battery.

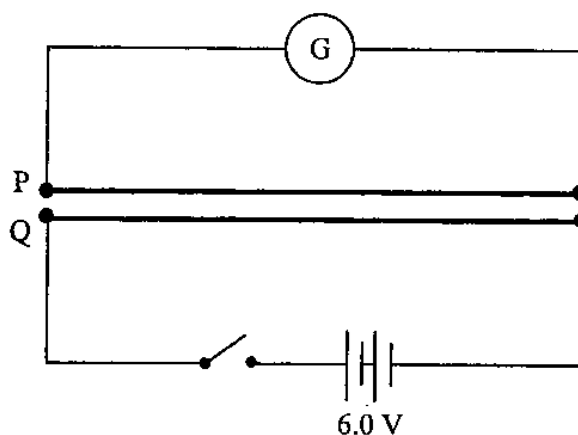


Fig. 4.3

State and explain any observations if the switch is closed.

[6]

Q14:NOV 2008 P5

- 1 (a) (i) State the **two** laws of electromagnetic induction.
- (ii) Fig. 1.1 shows how the magnetic flux, ϕ , linking a coil varies with time, t .

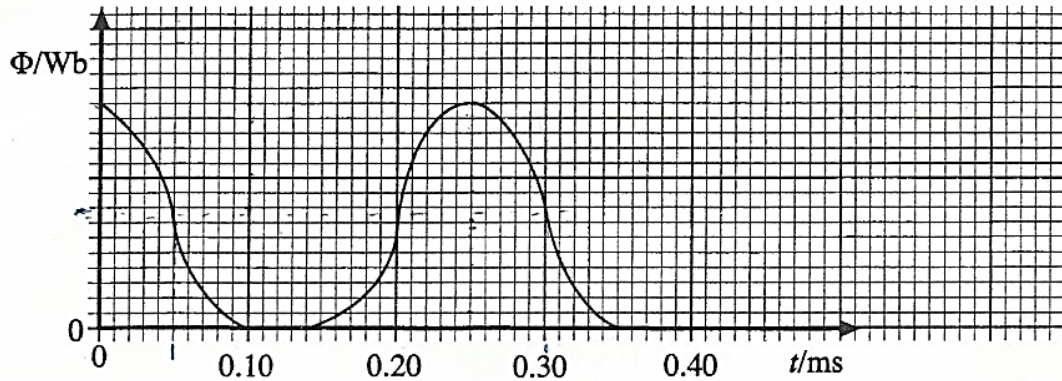


Fig. 1.1

- 1 Determine the frequency of the magnetic flux.
- 2 Sketch a graph to show how the induced e.m.f, E , varies with time
- 3 If the induced e.m.f. drives a current, I , through a resistor, sketch a graph of how the current varies with time. [8]

Q15:NOV 2008 P5

- 2 (a) Define the term *magnetic flux density*. [1]
- (b) A magnet is suspended from a forcemeter as shown in Fig. 2.1.

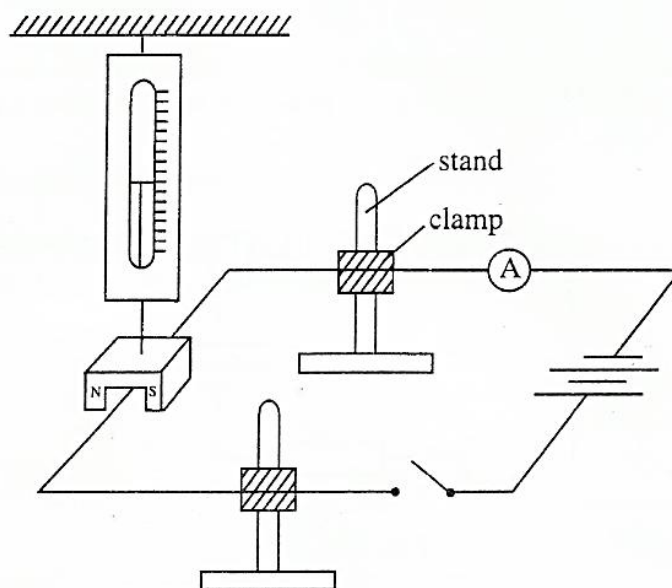


Fig. 2.1

4.5-4.6 ELECTROMAGNETISM AND INDUCTION P2, P3&P5 QUESTIONS

Compiled by : *R. Nuga*

A conductor of length 12.0 cm is held between the poles of the magnet so that it is at right angles to the magnetic field.

When the switch is open the forcemeter reads 2.36 N and when the switch is closed the forcemeter reads 2.29 N.

- (i) Explain why the forcemeter reading changed.
- (ii) Determine the value of the current, given that the magnetic flux density is 0.057 T.
- (iii) State and explain the effect of using alternating current.

[7]

- (c) An oscillating mass-spring system is **close** to an electromagnet as in Fig. 2.2.

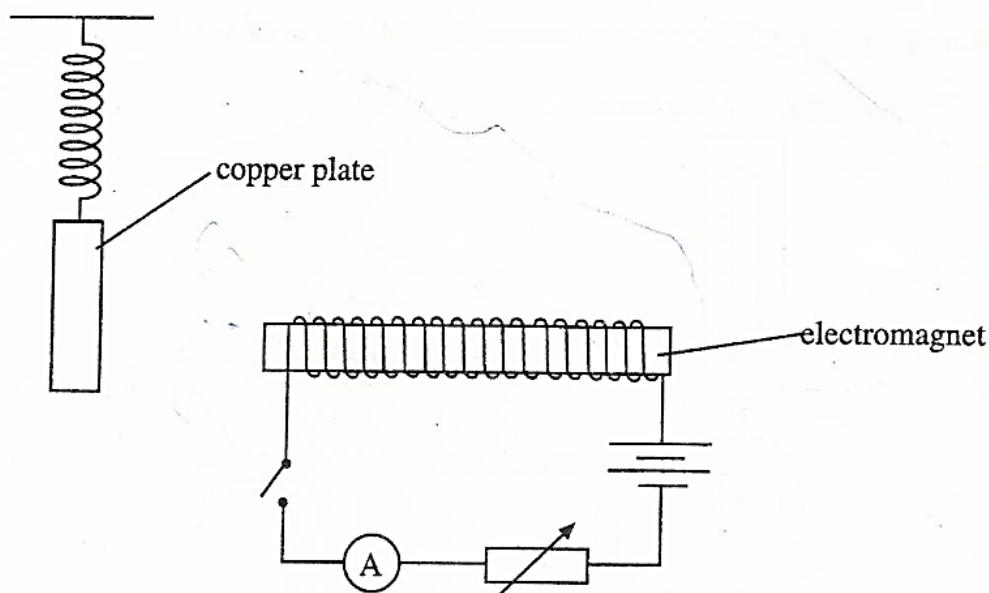


Fig. 2.2

Describe and explain the effect of closing the switch.

[4]

Q16:JUN 2009 P5

- 1 (a) (i) State Faraday's law.
- (ii) Two coils X and Y are placed close to each other as shown in Fig. 1.1.

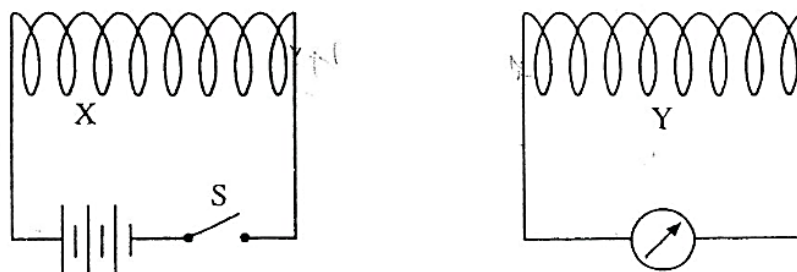


Fig. 1.1

Describe and explain the observations

1. on closing the switch S,
 2. when S is kept closed,
 3. when S is opened.
- (iii) Explain how the observations verify Lenz's law. [8]

Q17:JUN 2009 P5

- 3 (a) Oxygen atoms are ionised by the removal of electrons. The ions are accelerated in a vacuum and injected into a region where there is uniform magnetic and electric fields as shown in Fig. 3.1.

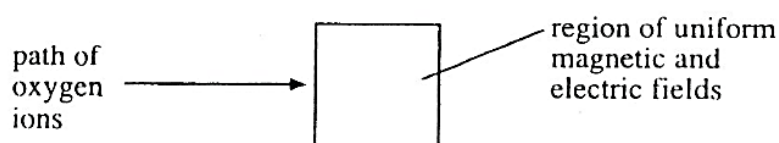


Fig. 3.1

- (i) Describe the conditions necessary for the ions to pass through the fields undeflected.
- (ii) If the speed of the ions as they reach the fields is 2.1×10^7 m/s and the magnetic flux density is 2.5×10^{-4} T, calculate the magnitude of the electric field strength so that the ions are not deflected in the region of the fields.
- (iii) State the change, if any, in the path of the ions in the two fields if the ionisation of oxygen resulted in the removal of one electron.

[6]

Q18:NOV 2009 P5

- 1 (c) (i) Define *magnetic flux* and *magnetic flux density*.
- (ii) When using a Hall probe, there is a Hall potential difference. Explain how this arises.
- (iii) The Hall potential difference across a chip of width 5.0 mm is $10\mu\text{V}$ when placed between the poles of a U-shaped magnet. The average drift speed of the charge carriers in the chip is $6.0 \times 10^{-1} \text{ mms}^{-1}$.

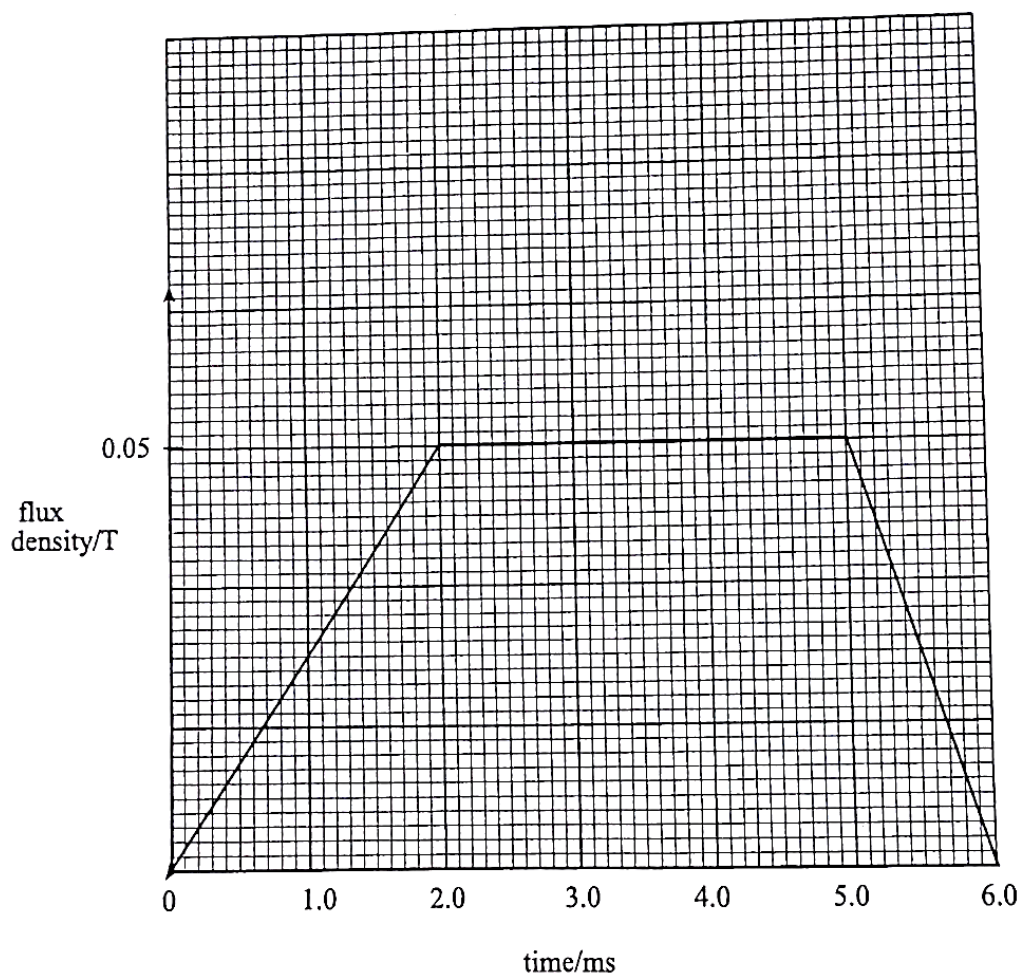
Determine the magnetic flux density between the poles of the magnet.

[8]

Q19:JUN 2012 P5

- 2 (a) State Faradays's Law of electromagnetic induction.
- (b) A coil of 600 turns of fine wire each of area 500 mm^2 is placed with its plane normal to a magnetic field. The ends of the coil are connected together. The magnitude of the magnetic flux density varies with time as shown in Fig. 2.1.

[1]



4.5-4.6 ELECTROMAGNETISM AND INDUCTION P2, P3&P5 QUESTIONS

Compiled by : R. Nuga

- (i) Calculate the maximum flux linking one turn of the coil.
- (ii) Determine the *emf* induced in the coil during the first 2.0 ms.
- (iii) Sketch a graph to show how the induced *emf* varies with time over the 6.0 ms.
- (iv) State three ways of increasing the magnitude of the induced e.m.f peak. [11]

Q20:NOV 2014 P5

- 1 (c) (i) State the difference between Lenz's and Faraday's laws of electromagnetic induction.
- (ii) Fig. 1.5 shows a petrol engine ignition system.

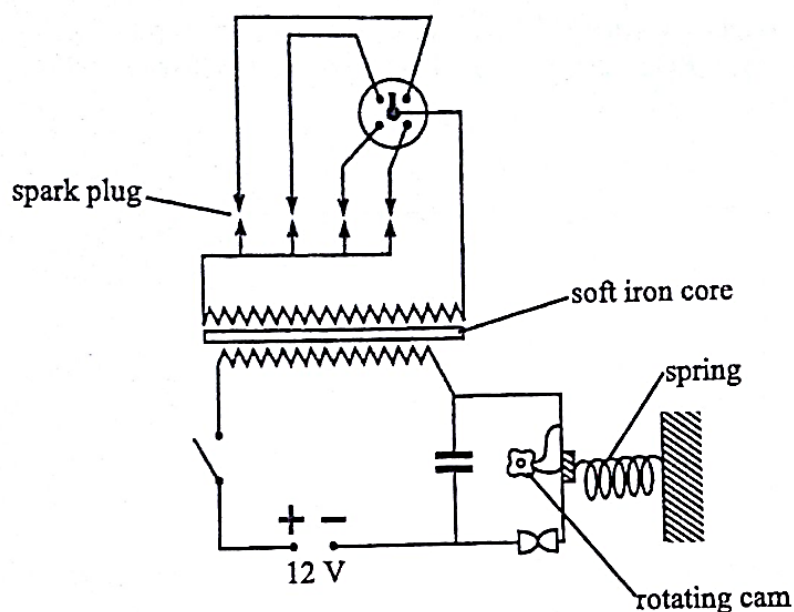


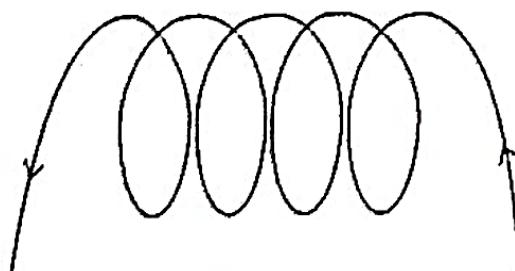
Fig. 1.5

Explain why the core is made of soft iron and its relevance to the circuit.

[4]

Q21:JUN 2015 P5

- 1 (a) Fig. 1.1 shows a solenoid carrying a current.



- (i) Sketch magnetic field flux patterns due to the solenoid.
- (ii) State how the magnetic field strength due to the solenoid in Fig.1.1 can be increased.

[4]

- (b) Faraday's law of electromagnetic induction is given by:

$$E = N \frac{d}{dt}(BA)$$

- (i) Name the physical quantity represented by the product BA .
- (ii) Fig. 1.2 shows a flat circular coil with 50 turns and a mean radius of 5.0 cm. It is turned over in 0.1 seconds.

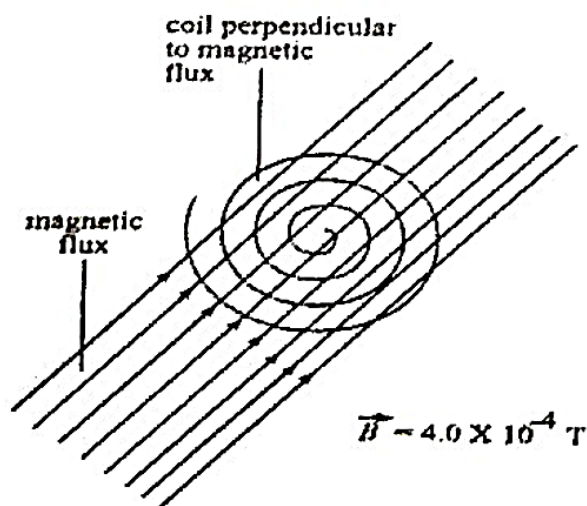


Fig.1.2

Calculate the mean e.m.f. induced when it is turned over in 0.1 seconds.

- (iii) Explain why there are several thousand turns in a coil of a practical generator.

[5]

Q22:NOV 2011 P5

- 2 (a) State Lenz's law of electromagnetic induction. [1]
- (b) In order to reduce fatigue to vehicle drivers, a designer has designed a braking system shown in Fig. 2.1.

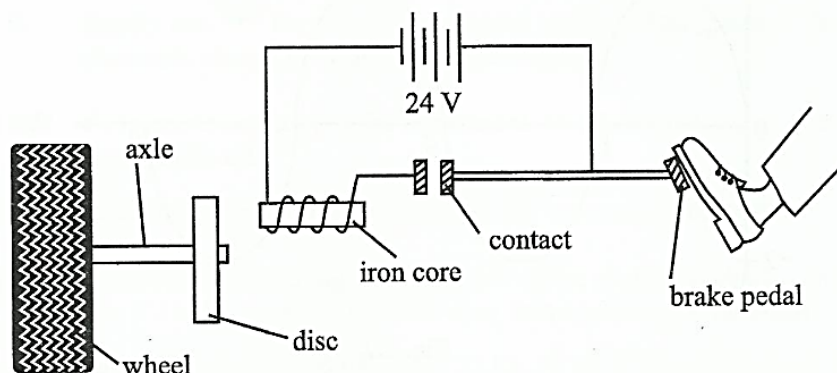


Fig. 2.1

- (i) Recommend a suitable material for the disc and give a reason for your choice.
- (ii) Describe and explain how the braking is achieved.
- (iii) State how the strength of the electromagnet can be increased.

[11]

Q23:JUN 2010 P5

- 2 (a) State Lenz's law of electromagnetic induction. [1]
- (b) Fig. 2.1 shows a coil fixed to a light trolley. The coil is given an initial push towards the wooden wall.

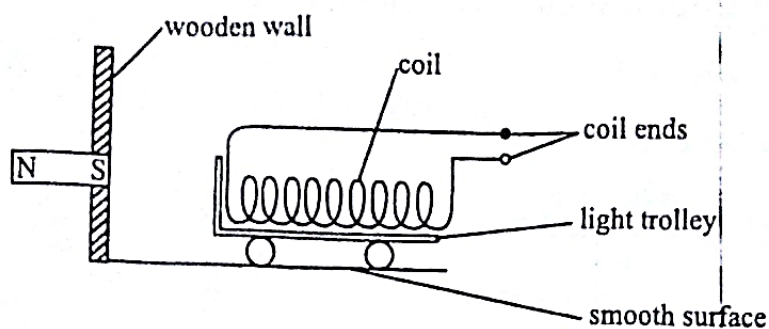


Fig. 2.1

- (i) Sketch the variation of the e.m.f. induced in the coil against time if the two ends of the coil are connected to a high resistance voltmeter.
- (ii) Explain the shape of your graph in (i)

- (iii) Describe and explain what would happen to the motion of the trolley if the experiment was repeated with the coil ends connected together.
- (iv) Calculate the e.m.f. induced in the coil if the rate of change of magnetic flux is 0.5 Wb/hr and the number of turns of coil is $5\,000$.

[11]

Q24:JUN 2011 P5

- 1 (a) (i) Sketch magnetic field patterns of a flat circular coil carrying current.
- (ii) Define *magnetic flux density*.
- (iii)

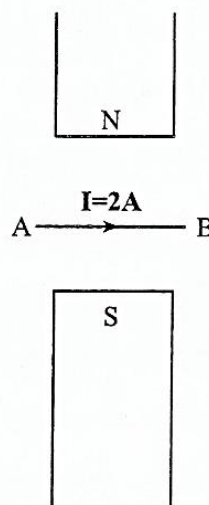


Fig. 1.1

Find the force per unit length acting on the wire AB in **Fig. 1.1** if the magnetic flux density is $2.4 \times 10^{-3} \text{ T}$.

[5]

- (b) **Fig. 1.2** shows electrons drifting from right to left through a block of a conductor. A magnetic flux density, $\mathbf{B}$, is applied upwards as shown.

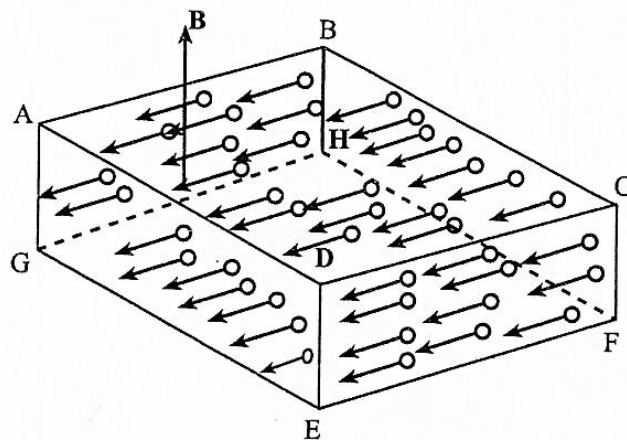


Fig. 1.2

Describe and explain the

- (i) effect of the magnetic field on the direction of the movement of electrons,
- (ii) movement of electrons if the magnetic field is applied parallel to their movement in (i).

[5]

Q25:JUN 2018 P5

- 1 (b) (i) State Faraday's Law of electromagnetic induction.
- (ii) A coil of 5 000 turns and cross-sectional area 10 cm^2 is rotated in the field at $100 \pi \text{ rads}^{-1}$.

The magnetic flux linkage $N\Phi$, in a magnetic density of 5 mT is given by the expression,

$$N\Phi = BAN\omega \sin \omega t.$$

1. Calculate the magnitude of the peak *emf* induced in the coil.
2. Sketch the graph of the variation of the magnitude of the induced *emf* with time, from $t = 0$ to $t = 0.04 \text{ s}$.

[7]

Q26:JUN 2017 P5

- 3 (a) (i) Write the equation for the force on a current-carrying conductor in a magnetic field.
- (ii) A wire of length 0.2 m carrying a current of 10 A, passes at right angles through a magnetic flux density of 0.15 T.
- Calculate the force acting on the wire.
- (iii) Two parallel current carrying conductors are held close to each other. State and explain what forces will act on the conductors when they carry current flowing in the same direction.

[6]

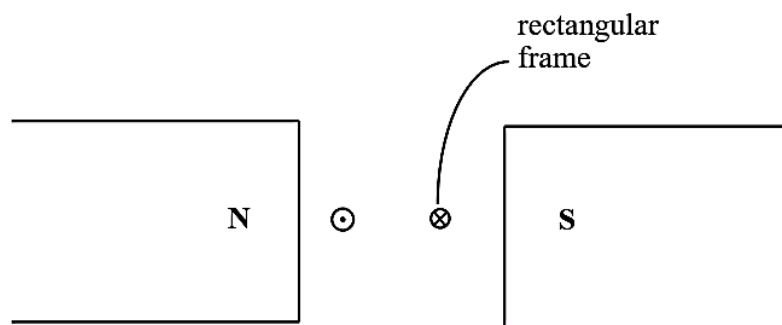
Q27:NOV 2018 P5

- 3 (c) (i) Define *magnetic flux linkage*.
- (ii) Use Lenz's and Faraday's laws of electromagnetic induction to show that the input to a transformer and the output are out of phase.

[3]

Q28:2018 SPEC P3

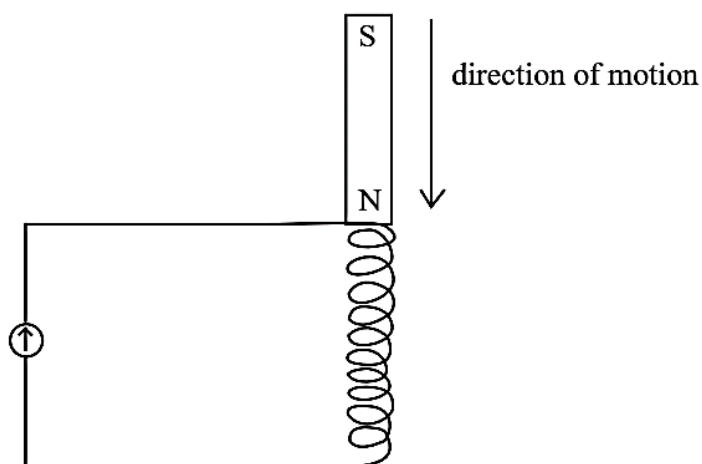
- 3 (a) (i) Explain what is meant by *magnetic field*.
- (ii) Sketch four lines to show the magnetic field around a long straight current carrying conductor.
- (iii) **Fig. 3.1** shows two parallel sides of a rectangular frame carrying a current. The frame is in a magnetic field.

**Fig. 3.1**

- Copy Fig. 3.1 and show the direction of the magnetic forces acting on the frame.
- State and explain the effect of the forces on the frame.

[8]

- (b) (i) **Fig. 3.2** shows a magnet plunging into a solenoid.

**Fig. 3.2**

With reference to **Fig. 3.2** explain how Lenz's law is an example of the principle of conservation of energy.

- (ii) **Fig. 3.3** shows the variation of the magnetic flux linking a coil of 200 turns.

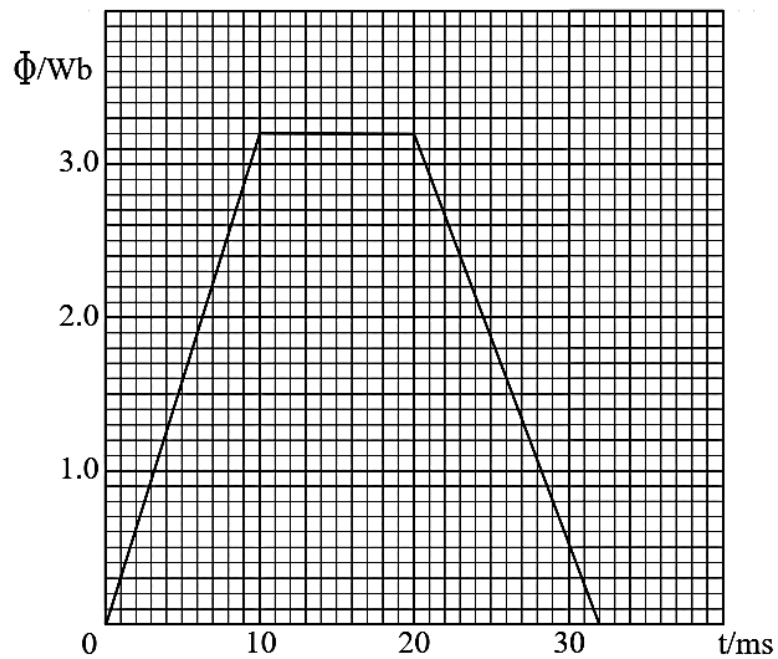


Fig. 3.3

- (ii) 1. Sketch a graph to show how the induced e.m.f. varies with time.
2. Suggest **two** ways by which the magnitude of the induced e.m.f. can be increased.

[8]